Exploring Aladin Sky Atlas through sound with SoniAladin

SoniAladin Installation guidelines

Download the full package from:
https://drive.google.com/drive/folders/1eKwrJuPZZoxo2iVf9166k-EStZnv6OKF?usp=sharing

Alternative download from Github: https://github.com/AuditoryVO/ICAD2026_Workshop

- **For mac users:**

1. SoniAladin runs in Python. Python is included in every macOS distribution package.
   a. Open *Terminal* and check your current python version. Write the following command and press enter:
      `$ python3 -V (or python3 -version)`
   b. If it is the first time you run python on your computer, you'll need to pre-install Xcode developer tools. Follow the installation wizard. Once Xcode is installed, check your Python version again.

2. SoniAladin requires several libraries which are listed in the provided *requirements.txt* file. To install them you will need first to install *pip*, the python standard package manager.
   a. On Terminal write the following command and press enter:
      `$ python3 -m ensurepip --upgrade`

   We will also create a virtual environment to run the project, preventing package version conflicts and keeping your Python installation clean.
   a. Install *venv*: `$ pip3 install virtualenv`
   b. Create the virtual environment: : `python3 -m virtualenv SoniAladin`
   c. Activate it: `$ source SoniAladin/bin/activate`
   d. Check the Python version in the new environment: `$ which python`
   e. Once finished you can deactivate the environment running: `$ deactivate`

3. Now you can install the dependencies running:
   `$ pip3 install -r requirements.txt`

   Important note: if Terminal returns an error related to the versions of the libraries you can delete the version reference in the requirements.txt file. For instance, replace "`numpy==2.2.6`" just by "`numpy`", save the file and run the command above again in Terminal.

4. Once all dependencies are installed you'll use the Driver IAC to send MIDI messages to the sound engine. To activate it:
    a. Open the *Audio and MIDI configuration* of your computer.
    b. Open the MIDI Studio at Window/Show MIDI Studio.
    c. Double click over Driver IAC.
    d. Activate the device selecting "*Device is online*".

5. We're almost ready! We will use GarageBand as your default sound engine.
    a. Open GarageBand
    b. Create a new file doing: *File/New/Empty Project*, then *MIDI Software Instrument*. The "Classic Electric Piano" is loaded by default. We will use this instrument to start, later you will have time to explore different sounds or different DAWs and synthesizers.

6. We will explore the Universe using Aladin Lite
    a. Open your web browser and go to: https://aladin.cds.unistra.fr/AladinLite/
    b. In the target field (automatically selected when Aladin lite is opened), write "Polaris" to go to the North Star.

7. Finally, run SoniAladin:
    a. On Terminal, navigate to the folder where you downloaded SoniAladin
    b. Run SoniAladin with:
    ```
    $ python3 SoniAladin.py
    ```
    c. Move to Aladin lite window (Command+Tab)
    d. Say "Music" to start the sonification.
    Important note: As SoniAladin uses screen captures and speech recognition to generate the sonifications, you'll need to allow access from Terminal to your microphone, keyboard and screen recording.
    e. Once all accesses are allowed you can control SoniAladin by voice and with the keyboard while navigating Aladin Lite Atlas.

8. You can modify SoniAladin code to customize your sonifications. You'll need a code editor such as Visual Studio (https://code.visualstudio.com/Download).

- **SoniAladin control (all OS):**

Say 'music' or press 'right arrow' to start the sonification. Say 'stop' or press 'left arrow' to stop the sonification. Say 'exit' or press'Esc' to finish.

- **For windows users:**

1. SoniAladin runs in Python. Install Python (current version at the moment is 3.13.12) in your computer from: https://www.python.org/downloads/windows/

2. SoniAladin requires several libraries which are listed in the provided *requirements.txt* file. To install them you will need first to install *pip*, the python standard package manager.
   a. On Terminal write the following command and press enter:
   `C:> py -m ensurepip --upgrade`

   We will also create a virtual environment to run the project. It prevents package version conflicts and keeps your Python installation clean.
   f. Install *venv* : `C:> pip install virtualenv`
   g. Create the virtual environment: : `C:> py -m virtualenv SoniAladin`
   h. Activate it: `C:> source SoniAladin/bin/activate`
   i. Check the Python version in the new environment: `C:> which python`
   j. Once finished you can deactivate the environment running: `C:> deactivate`

3. Now you can install the dependencies running:
   `C:> pip install -r requirements.txt`

   Important note: if Terminal returns an error related to the versions of the libraries you can delete the version reference in the requirements.txt file. For instance, replace "`numpy==2.2.6`" just by "`numpy`", save the file and run the command above again in Terminal.

4. SoniAladin automatically picks Microsoft GS Wavetable Synth, and plays directly. GS Wavetable is fully built in and reachable as a named MIDI output port, so Python MIDI output sounds with no extra configuration. We will use this instrument to start, later you will have time to explore different GM instruments or different DAWs and synthesizers.

   To change the GM Wavetable instruments, you can easily modify SoniAladin code. You'll need a code editor such as Visual Studio (https://code.visualstudio.com/Download) General MIDI instruments are assigned to programs (0-127). The following example changes channel 1 from piano (0) to harp (46):

```
channel = 1
program_number = 46
status = 0xC0 + (channel - 1)
midi_out.send_message([status, program_number])
```

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6. Finally, run SoniAladin:
    a. On Terminal, navigate to the folder where you downloaded SoniAladin
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```
C:> py SoniAladin.py
```

    c. Move to Aladin lite window (Command+Tab)
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    Important note: As SoniAladin uses screen captures and speech recognition to generate the sonifications, you’ll need to allow access from Terminal to your microphone, keyboard and screen recording.
    e. Once all accesses are allowed you can control SoniAladin by voice and with the keyboard while navigating Aladin Lite Atlas.

- **Resources for the exploration:**

IAU constellation maps
https://iauarchive.eso.org/public/themes/constellations/

Constellation	Abbreviation	English name
Ursa Major	UMa	The great bear
Ursa Minor	UMi	The little bear
Orion	Ori	The hunter
Cassiopeia	Cas	The Queen
Saggitarius	Sgr	The archer
Leo	Leo	The Lion
Taurus	Tau	The bull
Gemini	Gem	The twins
Andromeda	And	The princess
Pegasus	Peg	The winged horse

IAU star names
https://iauarchive.eso.org/public/themes/naming_stars/

IAU exoplanet names
https://iauarchive.eso.org/public/themes/naming_exoplanets/

Wiki list of galaxies
https://simple.wikipedia.org/wiki/List_of_galaxies